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date:06/02/2025

QUESTION 1

```
.data
n: .word 3
m: .word 4
s: .asciiz"\n sum of a & b
(a+b) = "
.text
.globl main
main:
la $t5,n
lw $t0,0($t5)
la $t5,m
lw $t1,0($t5)
li $v0,4
la $a0,s
syscall
add $t2, $t1, $t0
addi $v0, $0, 1
add $a0, $t2, $0
syscall
add $v0, $0, 10
syscall
```

Int Regs [10]

PC	=	4194396
EPC	=	0
Cause	=	0
BadVAddr	=	0
Status	=	805371664
HI	=	0
LO	=	0
R0	[r0]	= 0
R1	[at]	= 268500992
R2	[v0]	= 10
R3	[v1]	= 0
R4	[a0]	= 7
R5	[a1]	= 2147481620
R6	[a2]	= 2147481640
R7	[a3]	= 0
R8	[t0]	= 3
R9	[t1]	= 4
R10	[t2]	= 7
R11	[t3]	= 0
R12	[t4]	= 0
R13	[t5]	= 268500996
R14	[t6]	= 0
R15	[t7]	= 0
R16	[s0]	= 0
R17	[s1]	= 0
R18	[s2]	= 0
R19	[s3]	= 0
R20	[s4]	= 0

Text

Cons... — □

sum of a & b (a+b) = 7

QUESTION 2

```
.data
# Messages for displaying results
msg1: .asciiz "Before swap: \nFirst
number = "
msg2: .asciiz "\nSecond number = "
msg3: .asciiz "\n\nAfter swap: \nFirst
number = "
msg4: .asciiz "\nSecond number = "
newline: .asciiz "\n"
.text
.globl main
main:
li $s0, 10# First number = 10
li $s1, 20# Second number = 20
li $v0, 4
la $a0, msg1
syscall
li $v0, 1
move $a0, $s0
syscall
li $v0, 4
la $a0, msg2
syscall
li $v0, 1
move $a0, $s1
syscall
move $t0, $s0# temp = first
move $s0, $s1# first = second
move $s1, $t0# second = temp
li $v0, 4
la $a0, msg3
syscall
li $v0, 1
move $a0, $s0
syscall
li $v0, 4
la $a0, msg4
syscall
li $v0, 1
move $a0, $s1
syscall
li $v0, 4
```

Int Regs [10]		Cons...
PC	= 4194488	Before swap: First number = 10 Second number = 20 After swap: First number = 20 Second number = 10
EPC	= 0	
Cause	= 0	
BadVAddr	= 0	
Status	= 805371664	
HI	= 0	
LO	= 0	
R0 [r0]	= 0	
R1 [at]	= 268500992	
R2 [v0]	= 10	
R3 [v1]	= 0	
R4 [a0]	= 268501089	
R5 [a1]	= 2147481620	
R6 [a2]	= 2147481640	
R7 [a3]	= 0	
R8 [t0]	= 10	
R9 [t1]	= 0	
R10 [t2]	= 0	
R11 [t3]	= 0	
R12 [t4]	= 0	
R13 [t5]	= 0	
R14 [t6]	= 0	
R15 [t7]	= 0	
R16 [s0]	= 20	
R17 [s1]	= 10	
R18 [s2]	= 0	
R19 [s3]	= 0	
R20 [a4]	= 0	

```
la $a0, newline
syscall
li $v0, 10
syscall
```

QUESTION 3

```
.data
n: .word 50
m: .word 60
pro: .asciiz "\nproduct = "
.text
.globl main
main:
la $t5, n
lw $t0, 0($t5)
la $t5, m
lw $t1, 0($t5)
li $v0, 4
la $a0, pro
syscall
mul $t2, $t0, $t1
li $v0, 1
move $a0, $t2
syscall
add $v0, $0, 10
syscall
```

Int Regs [10]

PC	=	4194396
EPC	=	0
Cause	=	0
BadVAddr	=	0
Status	=	805371664
HI	=	0
LO	=	3000
R0	[r0]	= 0
R1	[at]	= 268500992
R2	[v0]	= 10
R3	[v1]	= 0
R4	[a0]	= 3000
R5	[a1]	= 2147481620
R6	[a2]	= 2147481640
R7	[a3]	= 0
R8	[t0]	= 50
R9	[t1]	= 60
R10	[t2]	= 3000
R11	[t3]	= 0
R12	[t4]	= 0
R13	[t5]	= 268500996
R14	[t6]	= 0
R15	[t7]	= 0
R16	[s0]	= 0
R17	[s1]	= 0
R18	[s2]	= 0
R19	[s3]	= 0
R20	[s4]	= 0

Cons...

product = 3000

QUESTION 4

```
.data
side: .word 8
area: .asciiz "\narea of
square = "
sid: .asciiz "\nlength of
edge = "
.text
.globl main
main:
la $t5,side
lw $t0,0($t5)
li $v0,4
la $a0,sid
syscall
li $v0,1
move $a0,$t0
syscall
li $v0,4
la $a0, area
syscall
mul $t1,$t0,$t0
li $v0, 1
move $a0, $t1
syscall
add $v0, $0,10
Syscall
```

FP Regs	Int Regs [10]	Data	Text
Int Regs [10]		Cons...	Text
PC	= 4194412		
EPC	= 0		
Cause	= 0		
BadVAddr	= 0		
Status	= 805371664		
HI	= 0		
LO	= 64		
R0	[r0] = 0		
R1	[at] = 268500992		
R2	[v0] = 10		
R3	[v1] = 0		
R4	[a0] = 64		
R5	[a1] = 2147481620		
R6	[a2] = 2147481640		
R7	[a3] = 0		
R8	[t0] = 8		
R9	[t1] = 64		
R10	[t2] = 0		
R11	[t3] = 0		
R12	[t4] = 0		
R13	[t5] = 268500992		
R14	[t6] = 0		
R15	[t7] = 0		
R16	[s0] = 0		
R17	[s1] = 0		
R18	[s2] = 0		
R19	[s3] = 0		
R20	[s4] = 0		

length of edge = 8
area of square = 64

QUESTION 5

```
.data
length: .word 8
breadth: .word 5
area: .asciiz "\narea of rectangle
= "
len: .asciiz "\nlength = "
bred: .asciiz "\nbreadth = "
.text
.globl main
main:
la $t5,length
lw $t0,0($t5)
la $t5,breadth
lw $t1,0($t5)
li $v0,4
la $a0,len
syscall
li $v0,1
move $a0,$t0
syscall
li $v0,4
la $a0,bred
syscall
li $v0,1
move $a0,$t1
syscall
li $v0,4
la $a0, area
syscall
mul $t2,$t0,$t1
li $v0, 1
move $a0, $t2
syscall
add $v0, $0,10
syscall
```

Int Regs [10]

```
PC      = 4194452
EPC     = 0
Cause   = 0
BadVAddr = 0
Status  = 805371664

HI      = 0
LO      = 40

R0 [r0] = 0
R1 [at] = 268500992
R2 [v0] = 10
R3 [v1] = 0
R4 [a0] = 40
R5 [a1] = 2147481620
R6 [a2] = 2147481640
R7 [a3] = 0
R8 [t0] = 8
R9 [t1] = 5
R10 [t2] = 40
R11 [t3] = 0
R12 [t4] = 0
R13 [t5] = 268500996
R14 [t6] = 0
R15 [t7] = 0
R16 [s0] = 0
R17 [s1] = 0
R18 [s2] = 0
R19 [s3] = 0
R20 [s4] = 0
```

Text

```
length = 8
breadth = 5
area of rectangle = 40
```